# Cell Division I

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Cell division plays an essential role in the growth, repair and reproduction of multicellular organisms. Cell division is the splitting of one (parent) cell into two identical (daughter) cells. There are two main types of cell division — *mitosis* and *meiosis*. Cells fall into two main classes *somatic* cells and *germ* cells.

**Somatic cells:** cells making up the tissues, organs, and parts of the body other than the germ (sex) cells.

Somatic cells (somatic, from the Greek, meaning 'body') are involved in growth and body maintenance. Mitosis (binary fission) produces new cells for growing organisms and also replaces cells, which have been damaged or have died. Many single-celled organisms reproduce *asexually* (without sex) using mitosis. Before cells divide in mitosis they grow, otherwise they would be smaller after each cell division.

**Diploid:** double, refers to two of each chromosome, a complete set. (Remember **D**iploid: **D**ouble.)

Somatic cells are usually *diploid* ($2n$), meaning they have a full set of chromosomes, one from each parent.

**Germ cells:** also called gametes, for example egg (ovum) cells or sperm cells.

Gametes (from the Greek, meaning 'wife') are set aside for reproductive purposes, but are not always used. These cells undergo meiosis, a cell division that produces *haploid* ($n$) cells. These cells require fusion (joining) with gametes from the opposite sex to restore the complement (full number).

**Haploid:** half, refers to half of one full set of chromosomes. (Remember **H**aploid: **H**alf.)

Human females are born with a lifetime's supply of gametes. When they reach maturity, one gamete (sometimes more) will be released monthly. Human males are not born with their gametes. Instead, males begin to form gametes at maturity and they are then produced as required.

The older a women is when she conceives, the greater the risk there is of her child suffering a genetic disorder. The risk becomes significant above the age of 35. For fathers, the risk of producing defective gametes also increases with age, but only after the age of 50 does this pose a problem.

1. What type of cell division produces cells used in the growth and repair of the body?
2. What type of cell division produces sperm and ova?
3. Give another name for a 'gamete'.
4. What happens to the number of chromosomes in mitosis?
5. Why do cells undergoing mitosis need to grow before they divide?
6. What is the difference between 'diploid' cells and 'haploid' cells?
7. What makes the gametes of a mature female more likely to be defective than the gametes of a man of the same age?
8. 'All living things grow by cell division.' True or false?
9. 'Binary fission means two cells joining.' True or false?
10. 'Ovulation is the process by which females release eggs every month.' True or false?
11. 'Human males are born with a complete set of gametes.' True or false?
12. '*Daughter* cells are always female.' True or false?



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